

System & Network Security Implementation Concepts (CS0-003)

Introduction

Objective

Welcome to the **System & Network Security Implementation Concepts** practice lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Overview

Learning Outcomes:

In this module, you will complete the following exercises:

- Exercise 1 – Log Collection with Splunk
- Exercise 2 – Encrypting Sensitive Data
- Exercise 3 – Enable Multifactor Authentication

After completing these modules, you should be able to:

- Collect Logs from Devices on the Network using the Splunk Enterprise Application
- Encrypt a Local Drive using BitLocker Drive Encryption
- Enable Multifactor Authentication

Exam Objectives:

The following exam objectives are covered in this module:

- 1.1 Explain the importance of system and network architecture concepts in security operations.

Exercise 1 – Log Collection with Splunk

The Splunk Enterprise application can be used to collect logs of devices on the network. Using the application, a centralized point of all the logs of the devices can be monitored.

In this exercise, the Splunk Enterprise application will be installed and configured to collect the logs of devices on the network.

Learning Outcomes

After completing this exercise, you should be able to:

- Install and Configure the Splunk Enterprise Application
- Collect Logs from Devices on the Network

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIDM01** – Windows Server 2022 – Domain Member Server
- **ACIWIN11** – Windows 11 – Domain Member Workstation

Task 1 – Install and Configure the Splunk Enterprise Application

The Splunk Enterprise application collects logs from devices on the network. In this task, the Splunk Enterprise application will be installed and configured.

1

Click the **Start** button.



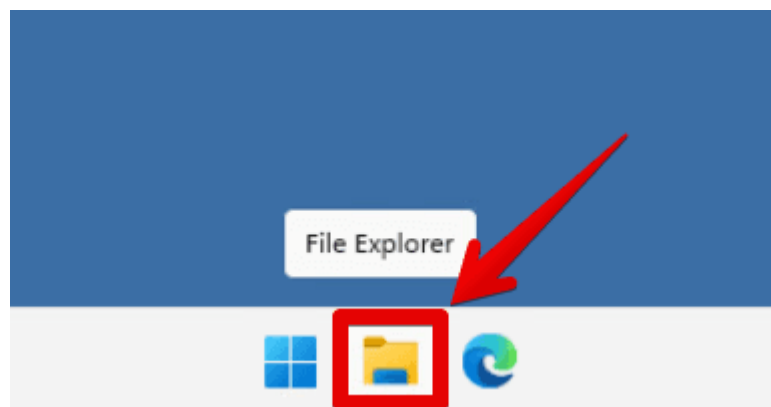
2 **Connect** to ACIWIN11.

Click **Send Ctrl+Alt+Delete** button on the top of the screen.

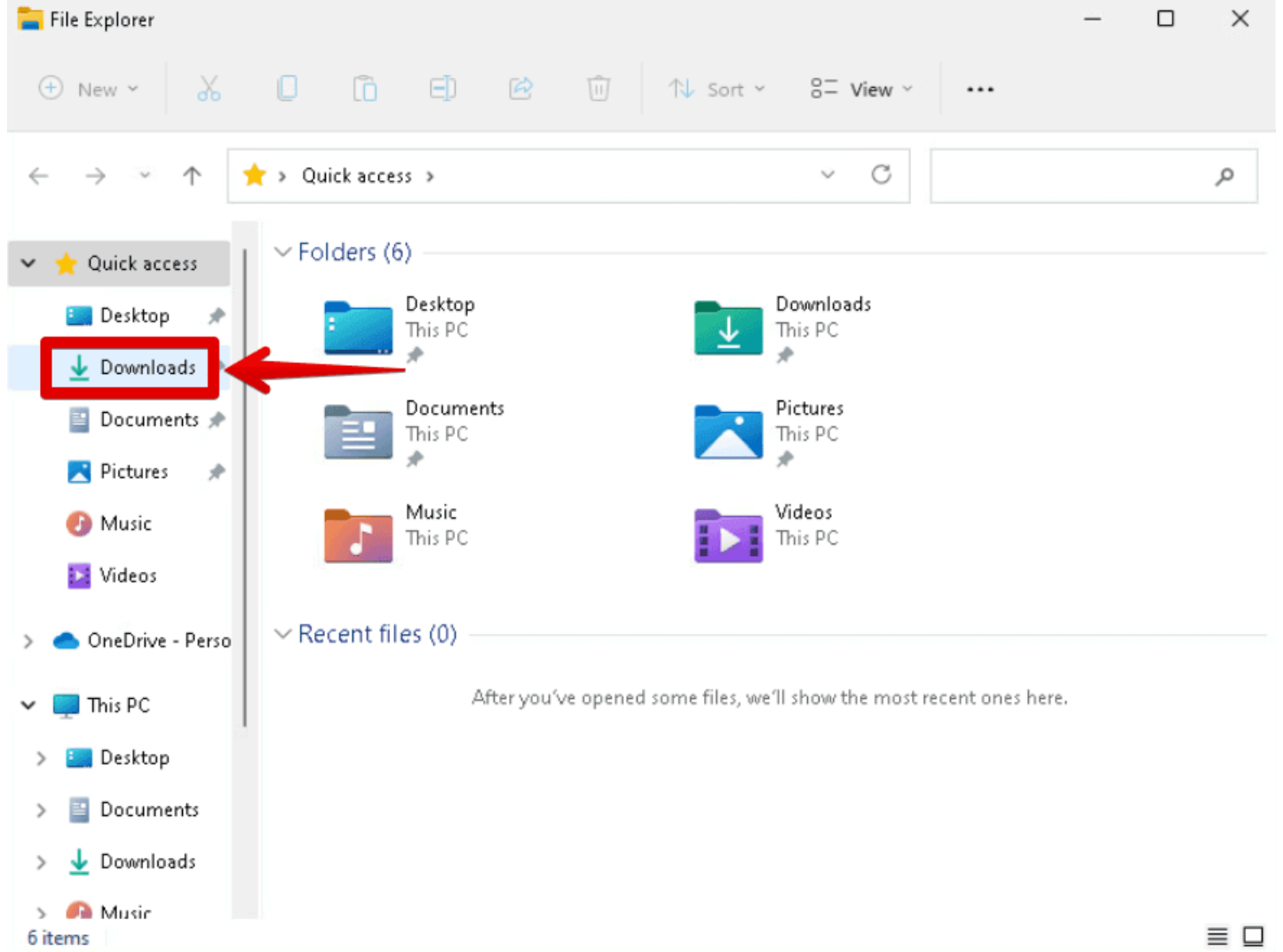
Username: ACIPLAB\Administrator

Password: Passw0rd

From the **Taskbar**, **open** File Explorer.

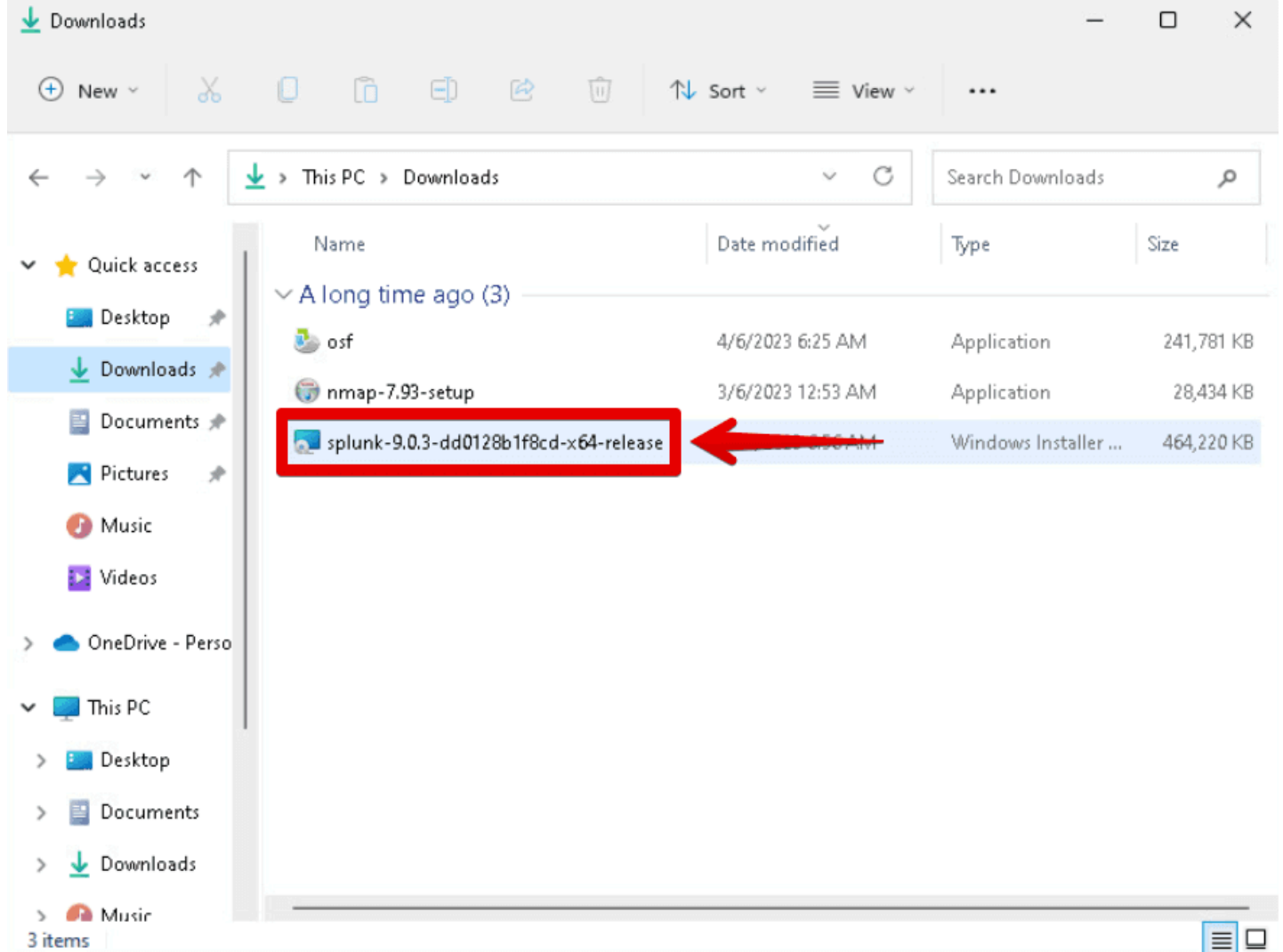


3 In **File Explorer**, **click** the **Downloads** folder.



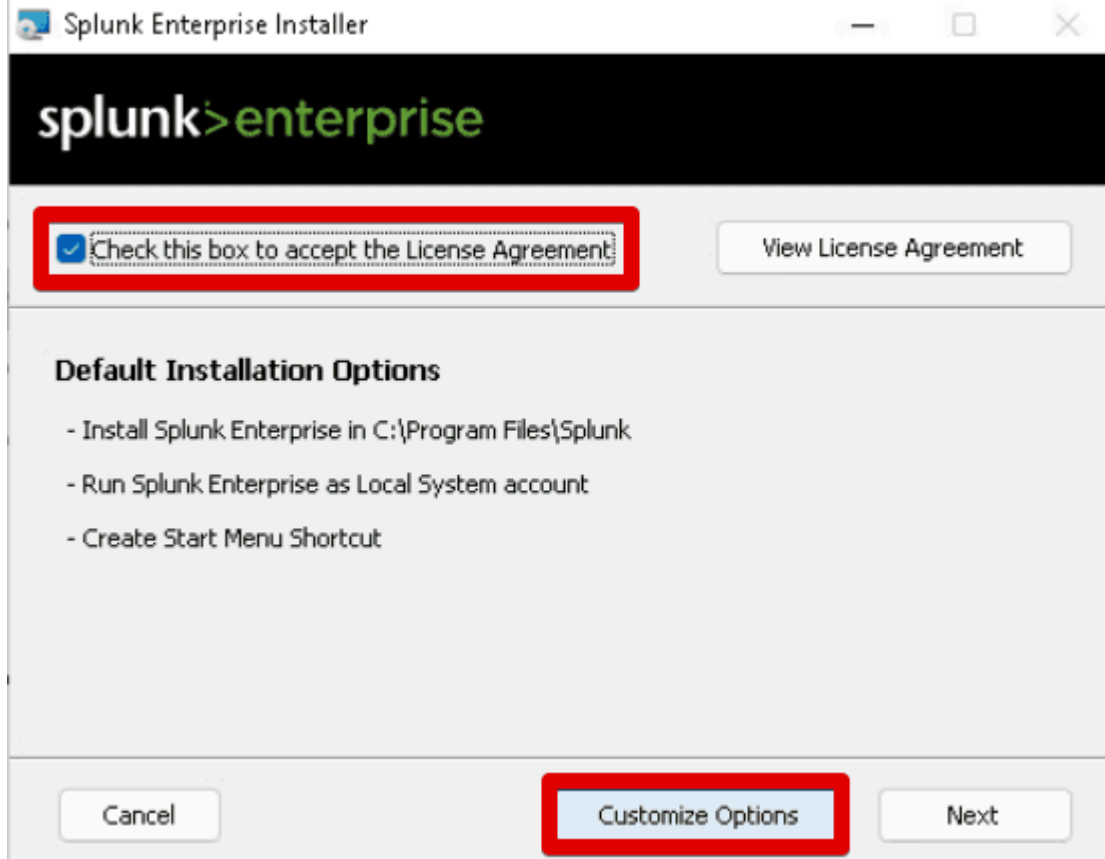
4

In File Explorer in the Downloads folder, **double-click** the `splunk-9.0.3-dd` installation file.



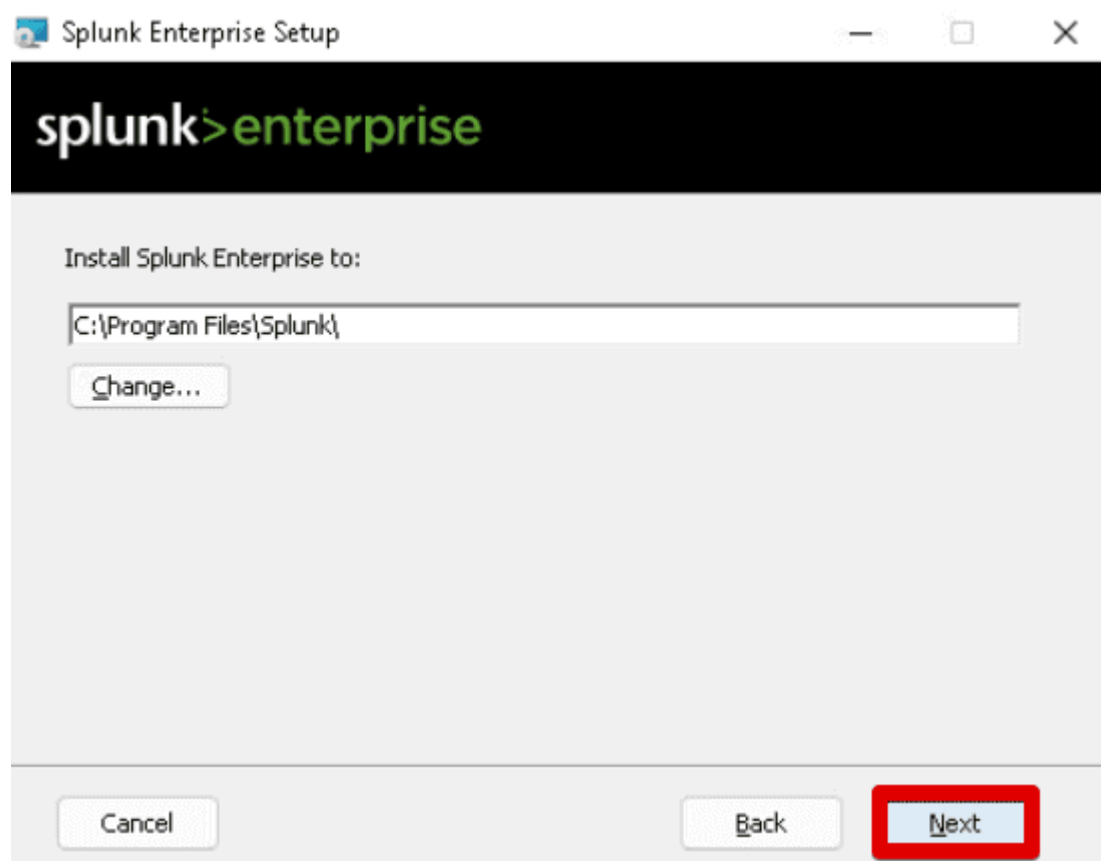
- 5 On the **Splunk Enterprise Installer** pop-up window, **tick** the **Check this box to accept the License Agreement** tick box.

Click **Customize Options**.



6

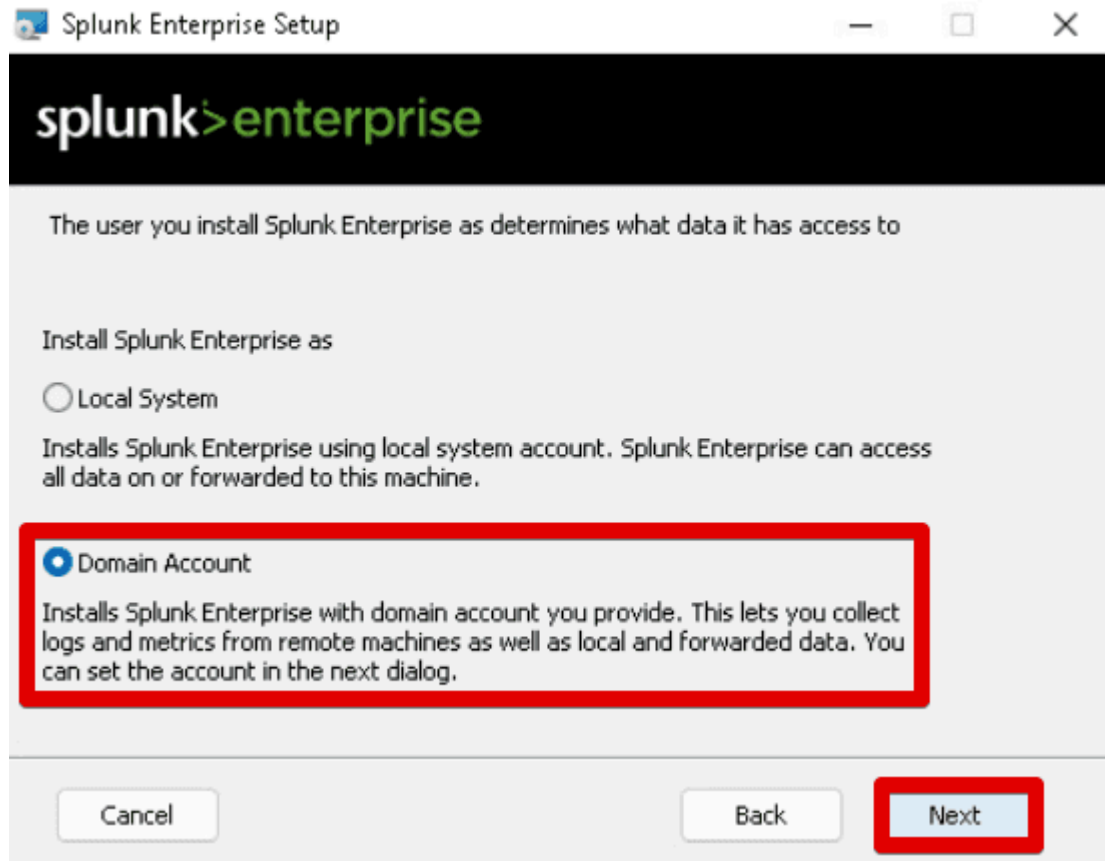
Click **Next** on the **Install Splunk Enterprise** screen.



7

Select **Domain Account** on the **the Install Splunk Enterprise** screen.

Click **Next**.



8

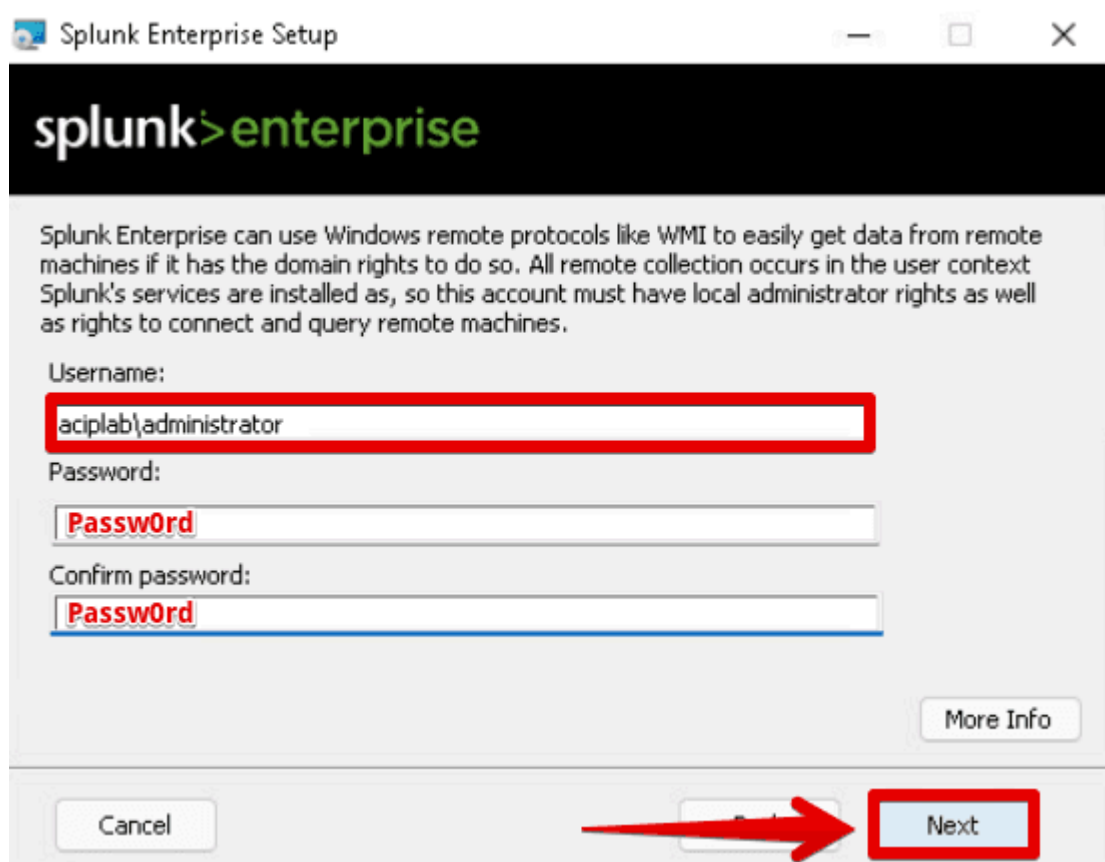
Enter the following details on the Splunk Enterprise Setup screen:

Username:

Password:

Confirm password:

Click Next.





Note: Using the **Domain Administrator account** for installation of **Splunk Enterprise** will ensure the application will be able to collect logs from devices on the network

9

Enter the following details on the **Splunk Enterprise Setup** screen:

Username: `splunkadmin`

Password: `Passw0rd`

Confirm password: `Passw0rd`

Click **Next**.

Splunk Enterprise Setup

splunk>enterprise

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:
`splunkadmin`

Password:
`Passw0rd`

Confirm password:
`Passw0rd`

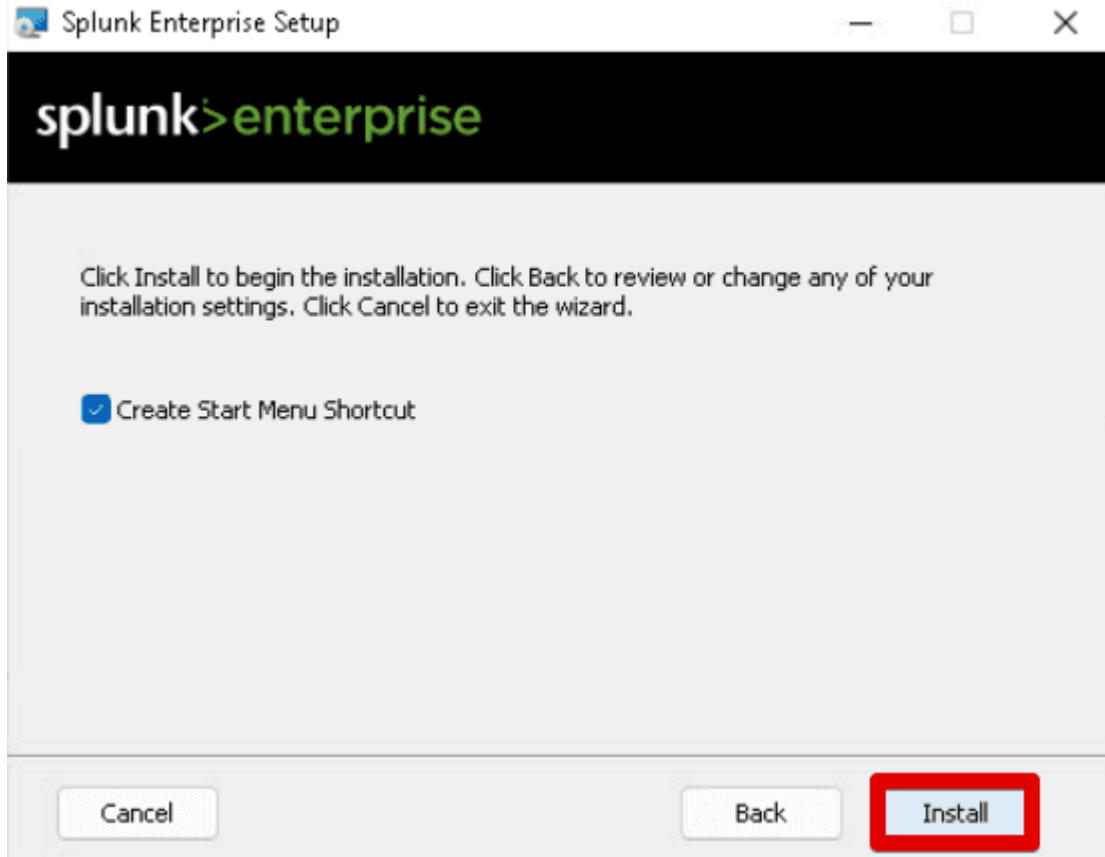
Cancel Next



Note: A local account with a password needs to be created that will be used to log into the Splunk Enterprise web application.

10

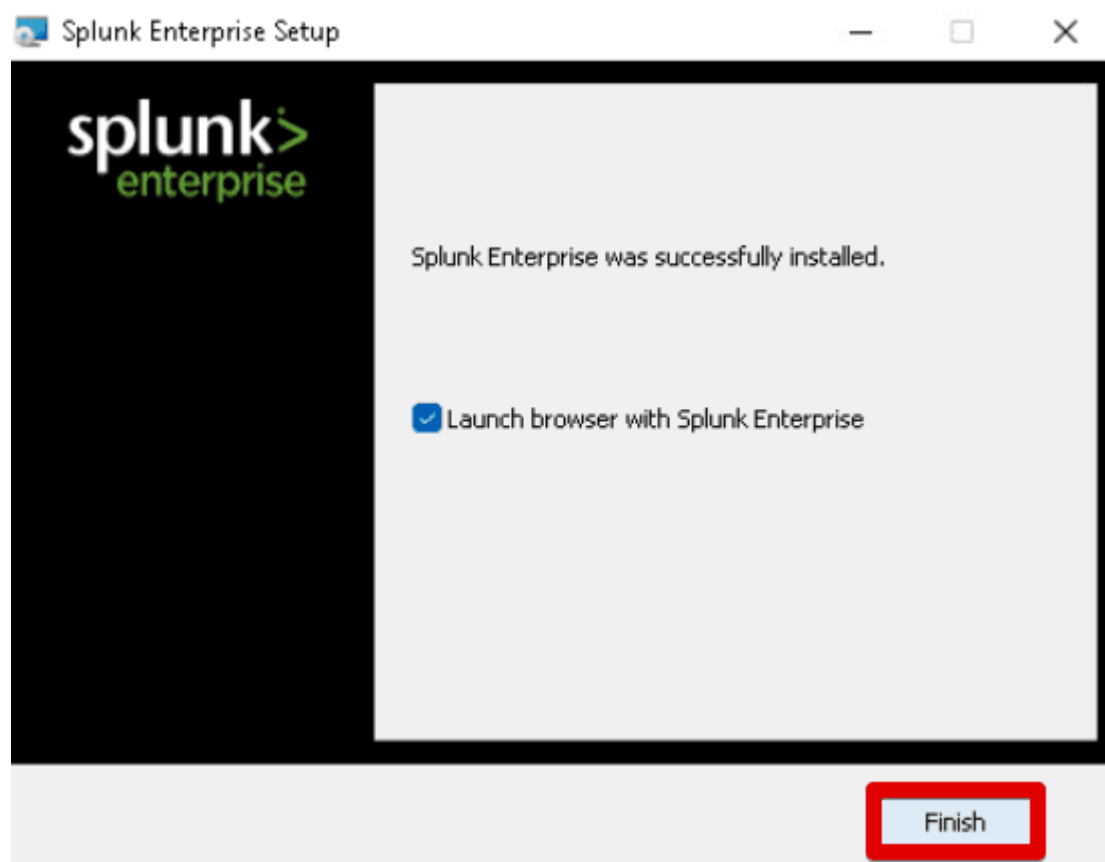
Click **Install** on the **Splunk Enterprise Setup** screen.



Note: The installation will take a couple of minutes to complete.

11

Click **Finish** on the **Splunk Enterprise Setup** screen.





Note: The **Splunk Enterprise** web application will open automatically in the Microsoft Edge browser.

- 12 If the **Microsoft Edge browser window** does not prompt up, **click** on **Microsoft Edge** from the **taskbar** and **enter** the **following credentials**:

Username:

Password:

Click **Sign In**.

splunk>enterprise

splunkadmin Passw0rd Sign In

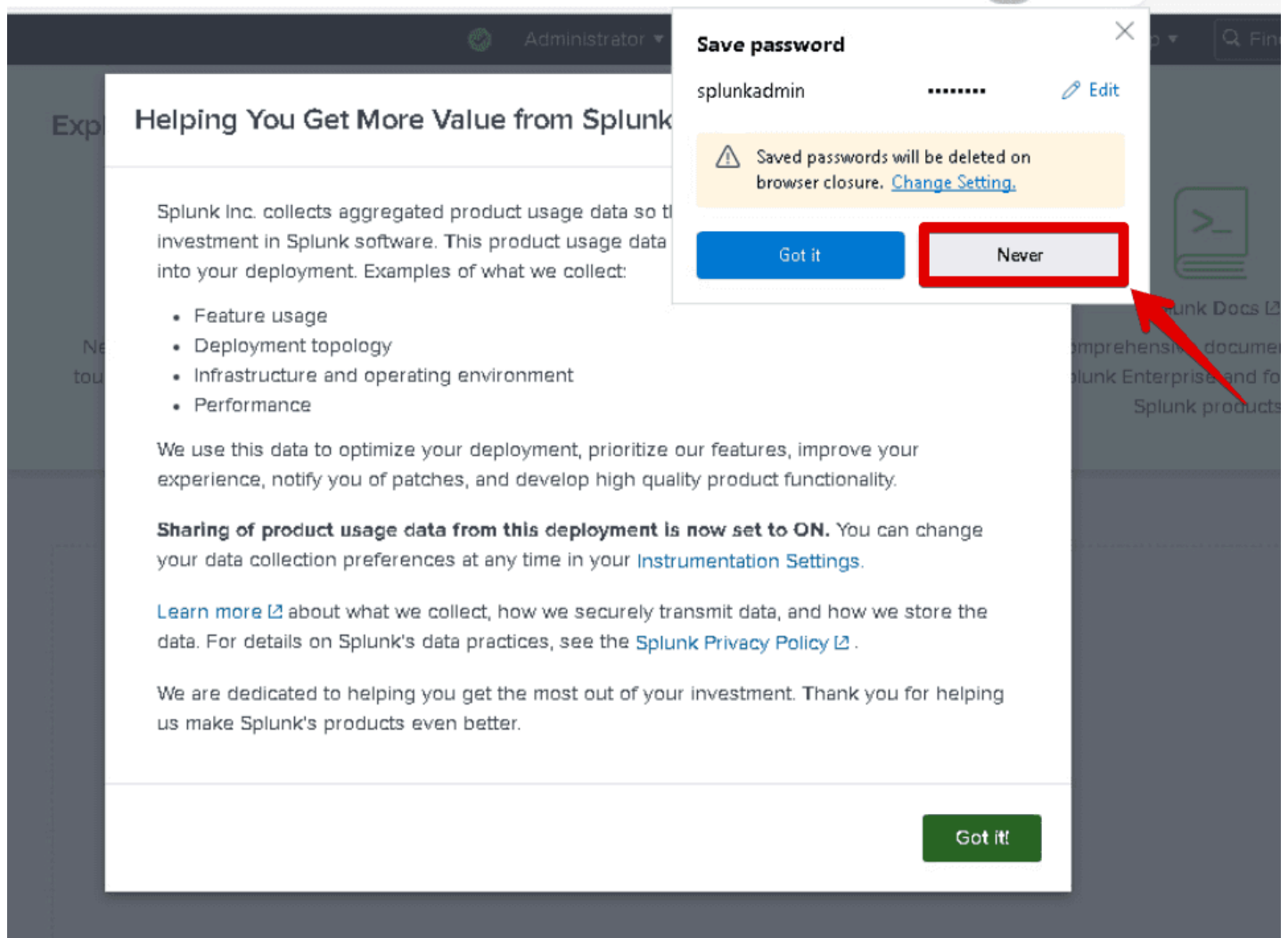
First time signing in?

If you installed this instance, use the username and password you created at installation. Otherwise, use the username and password that your Splunk administrator gave you. If you've forgotten your username or password, please contact your Splunk administrator.

username admin
password The password you created when you installed this instance

First time signing in?

- 13 **Click** **Never** on the **Save password pop-up window**.



14

Click **Got it** on the **Helping You Get More Value from Splunk Software** pop-up window.

Exp Helping You Get More Value from Splunk Software

Splunk Inc. collects aggregated product usage data so that we can enhance the value of your investment in Splunk software. This product usage data does not include any data you ingest into your deployment. Examples of what we collect:

- Feature usage
- Deployment topology
- Infrastructure and operating environment
- Performance

We use this data to optimize your deployment, prioritize our features, improve your experience, notify you of patches, and develop high quality product functionality.

Sharing of product usage data from this deployment is now set to ON. You can change your data collection preferences at any time in your [Instrumentation Settings](#).

[Learn more](#) [about what we collect, how we securely transmit data, and how we store the data.](#) For details on Splunk's data practices, see the [Splunk Privacy Policy](#).

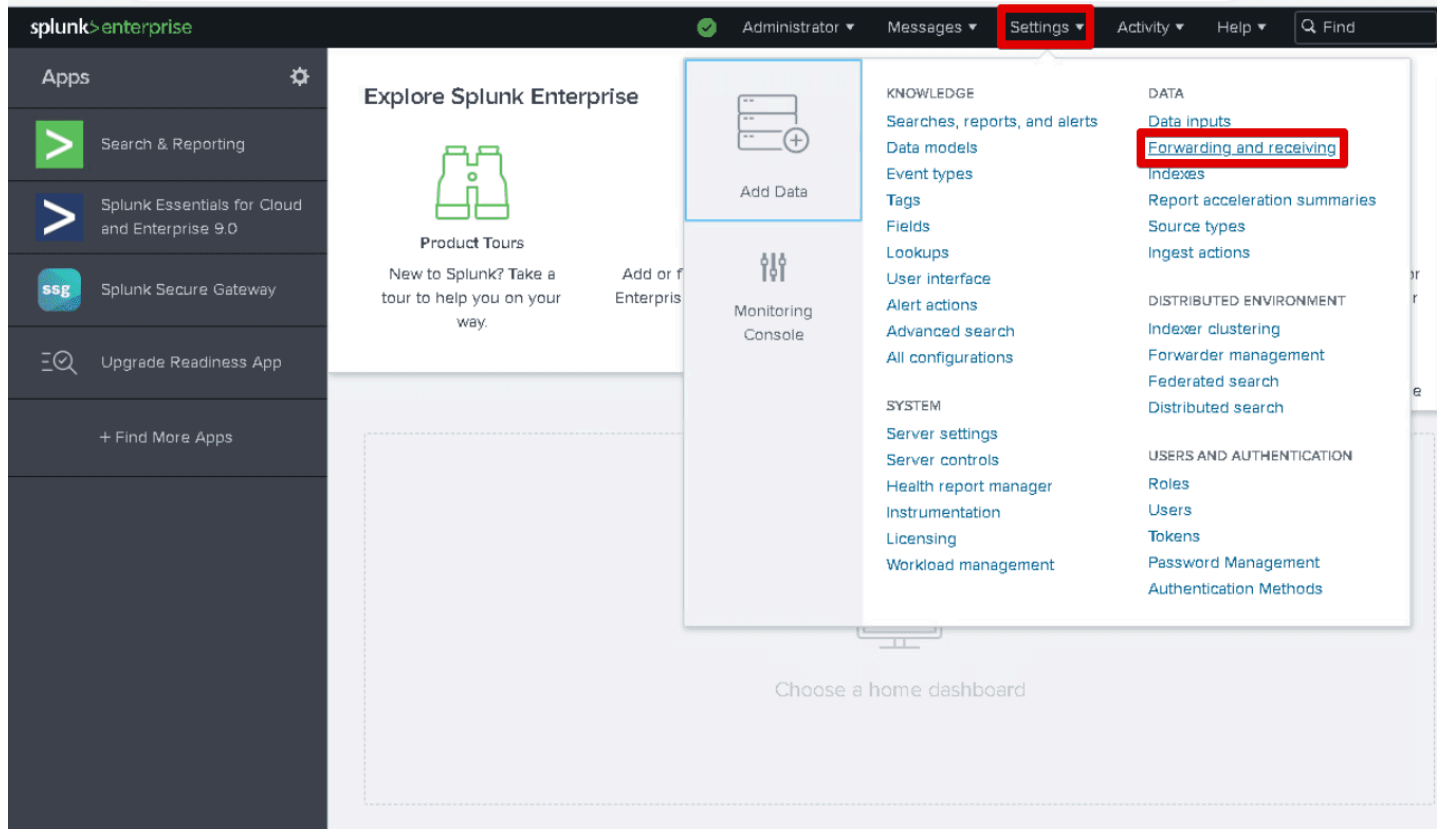
We are dedicated to helping you get the most out of your investment. Thank you for helping us make Splunk's products even better.



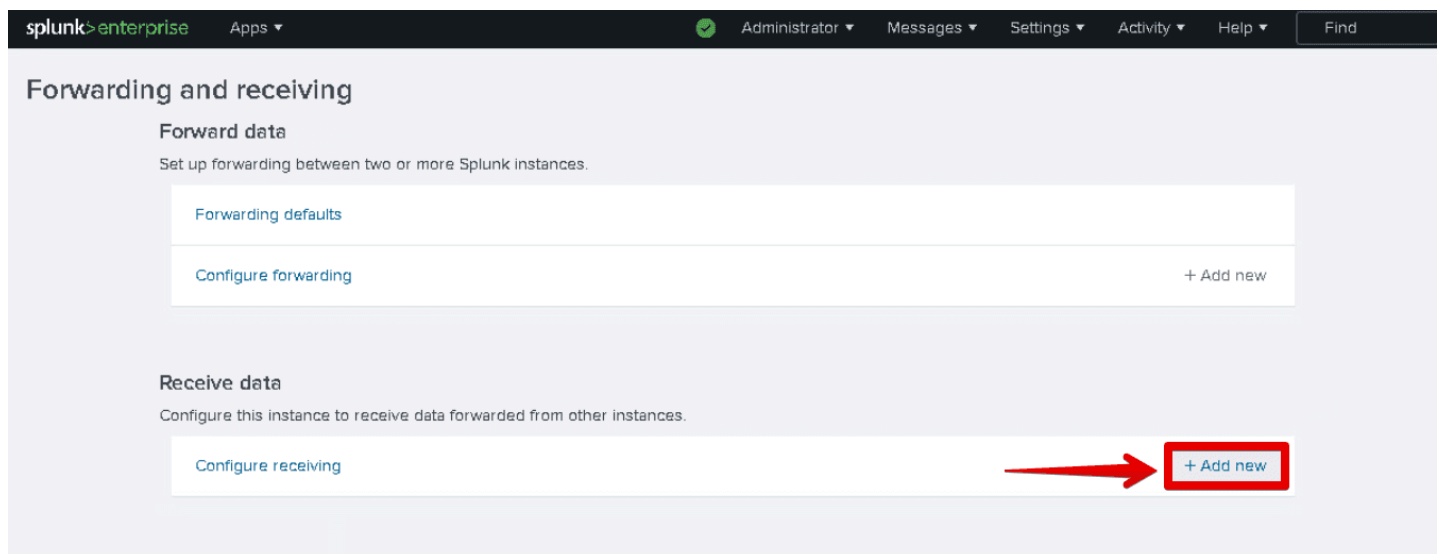
15

In the **Splunk Enterprise web application**, **click** **Settings**.

In the **fly-out window**, in the **DATA pane**, **select** **Forwarding and receiving**.



- 16 On the **Forwarding and receiving** window, **click** **+ADD new** in the **Receive data** pane.



- 17 On the **Add new** window, **enter** the **following** in the **Listen on this port** field:

9997

Click **Save**.

splunk>enterprise Apps ▾ Administrator ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find

Add new

Forwarding and receiving > Receive data > Add new

Configure receiving

Set up this Splunk instance to receive data from forwarder(s).

Listen on this port *

For example, 9997 will receive data on TCP port 9997.

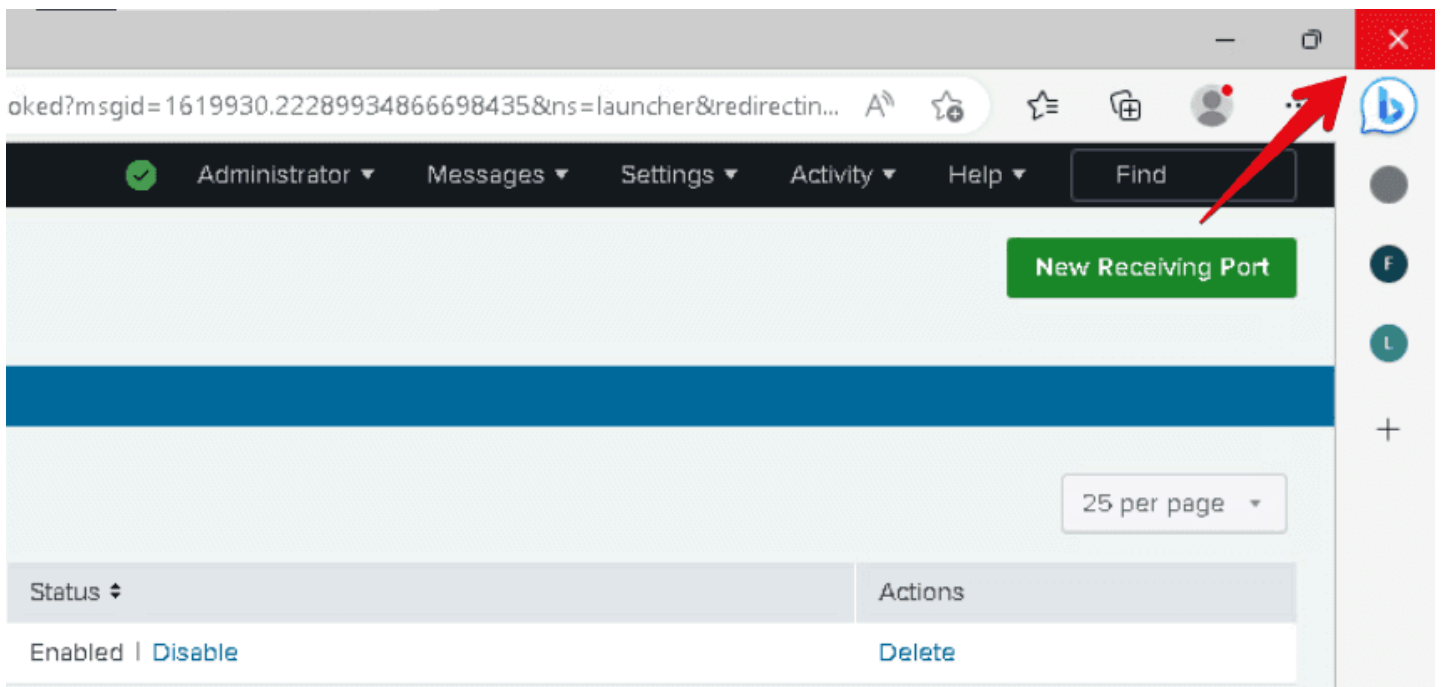
Cancel Save



Note: The **Splunk Enterprise application** was configured to listen for network traffic. The **Windows Firewall** needs to be configured to ensure the traffic can be received on the port.

18

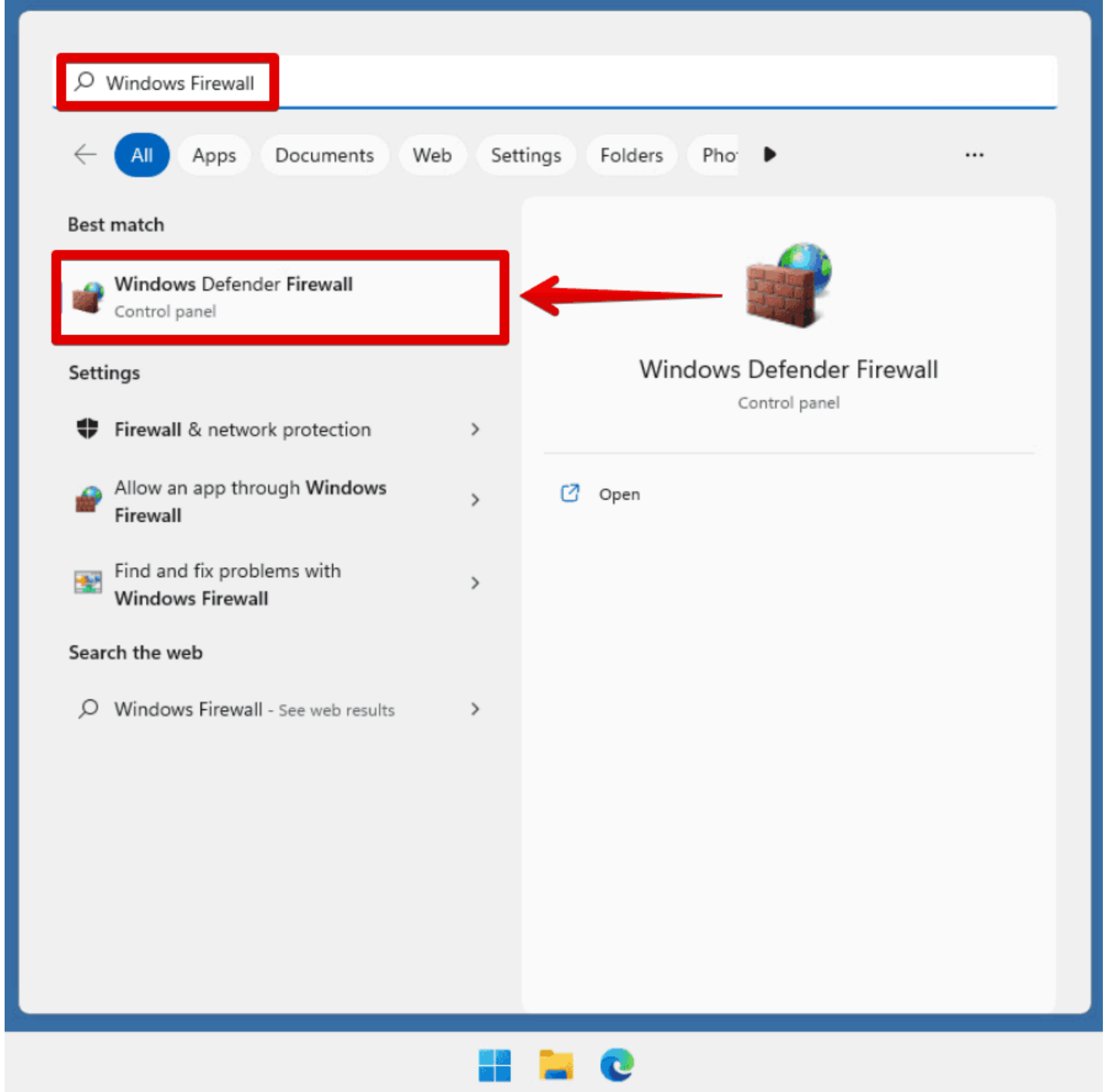
Close the **Microsoft Edge** browser.



19

Click **Start**, **type** the **following**, and **press** **Enter**:

Windows Firewall

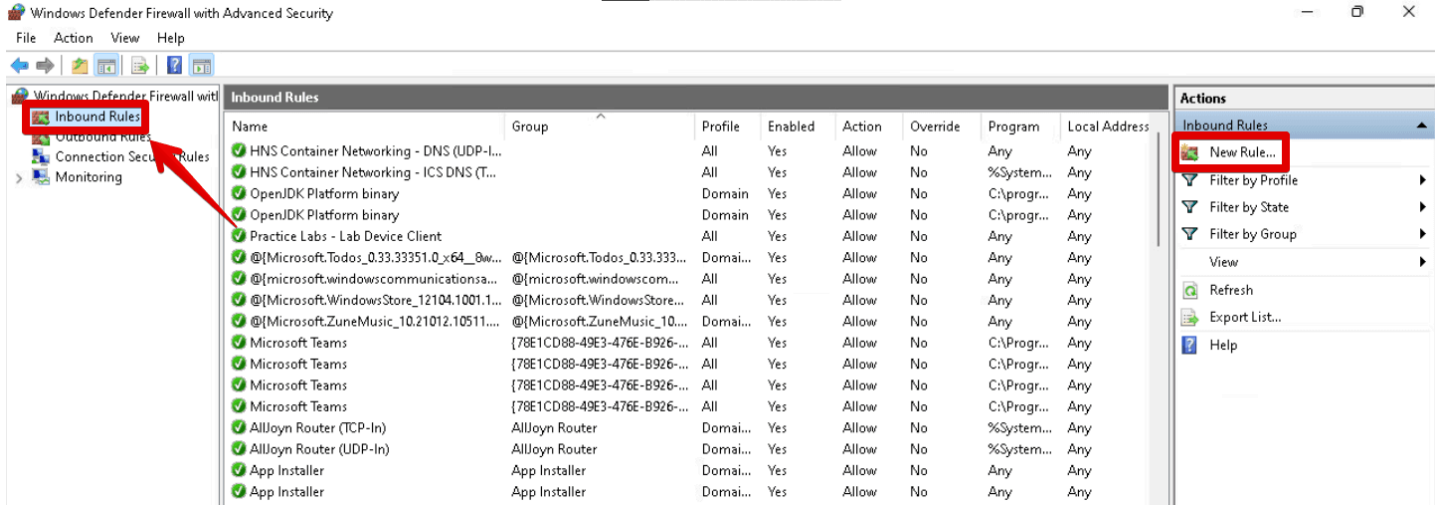


20

On the **Windows Defender Firewall** window, **click** **Advanced settings**.

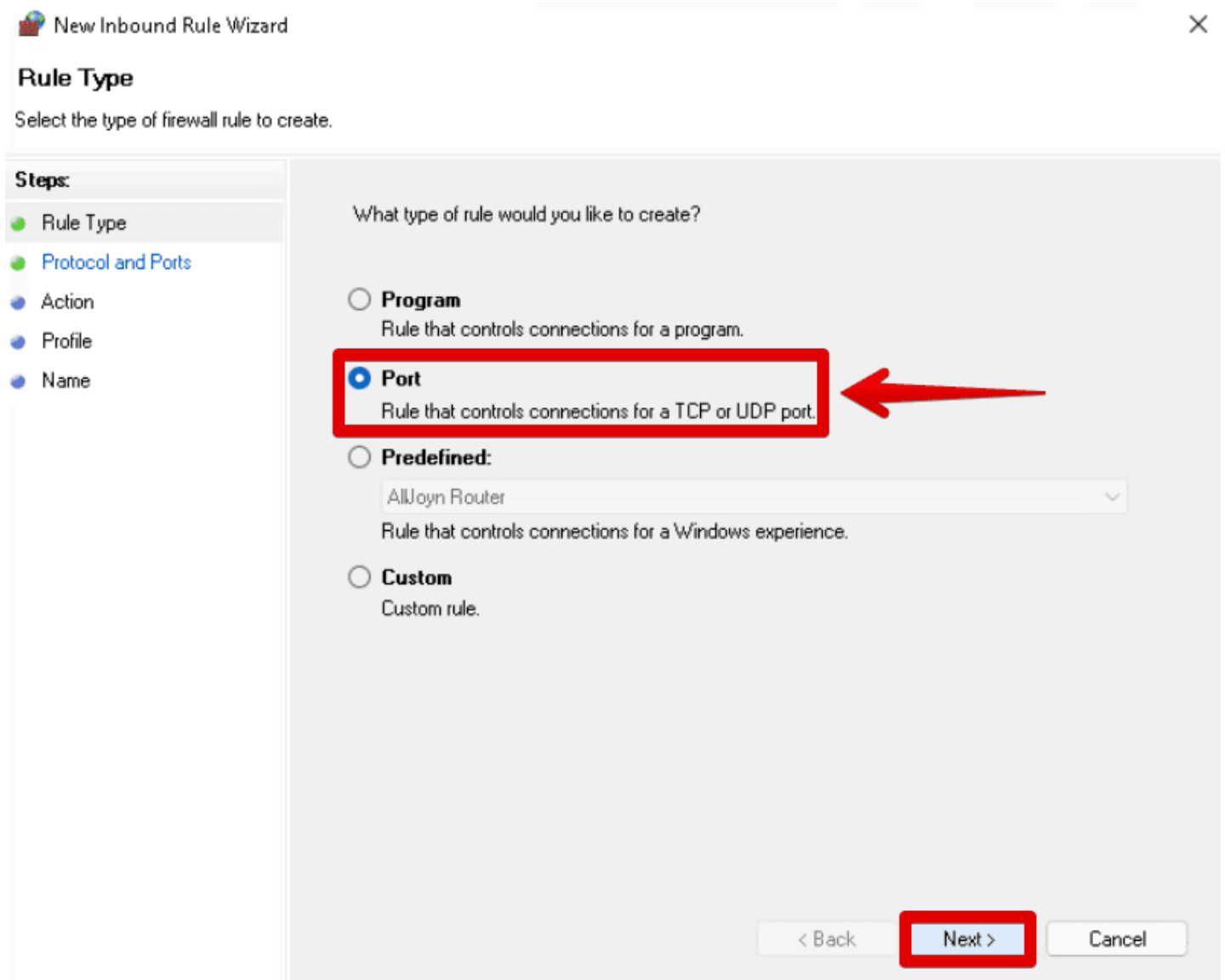
21

On **Windows Defender Firewall with Advanced Security**, **select** **Inbound Rules** and **click** **New Rule** in the right pane.



22 Select **Port** on the **Rule Type** window.

Click **Next**.



23 Enter the **following** in the **Specific local ports** field:

8089, 9997

Click **Next**.

New Inbound Rule Wizard ✕

Protocol and Ports

Specify the protocols and ports to which this rule applies.

Steps:

- Rule Type
- Protocol and Ports**
- Action
- Profile
- Name

Does this rule apply to TCP or UDP?

☒ **TCP**

☐ **UDP**

Does this rule apply to all local ports or specific local ports?

☐ **All local ports**

☒ **Specific local ports:**

Example: 80, 443, 5000-5010

24

Click **Next** on the **Action** and **Profile** windows.

Profile

Specify the profiles for which this rule applies.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

When does this rule apply?

☒ **Domain**

Applies when a computer is connected to its corporate domain.

☒ **Private**

Applies when a computer is connected to a private network location, such as a home or work place.

☒ **Public**

Applies when a computer is connected to a public network location.

< Back

Next >

Cancel

25

Enter the following in the Name field and **click** Finish.

Splunk

New Inbound Rule Wizard

Name

Specify the name and description of this rule.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name**

Name: Splunk

Description (optional):

< Back Finish Cancel

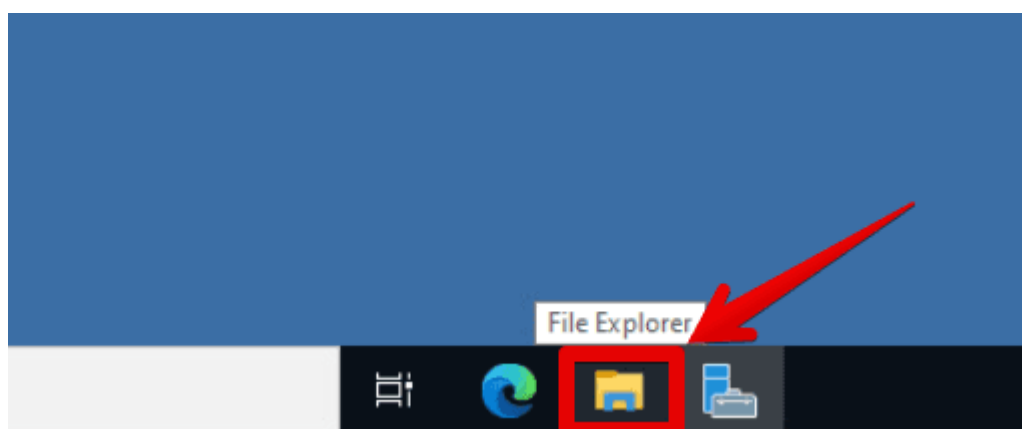
Task 2 – Configuring a Splunk Client

To ensure the devices on the network can send log activity, the device needs to be configured to facilitate this.

In this task, a client device will be configured to forward its logs to the Splunk Enterprise application.

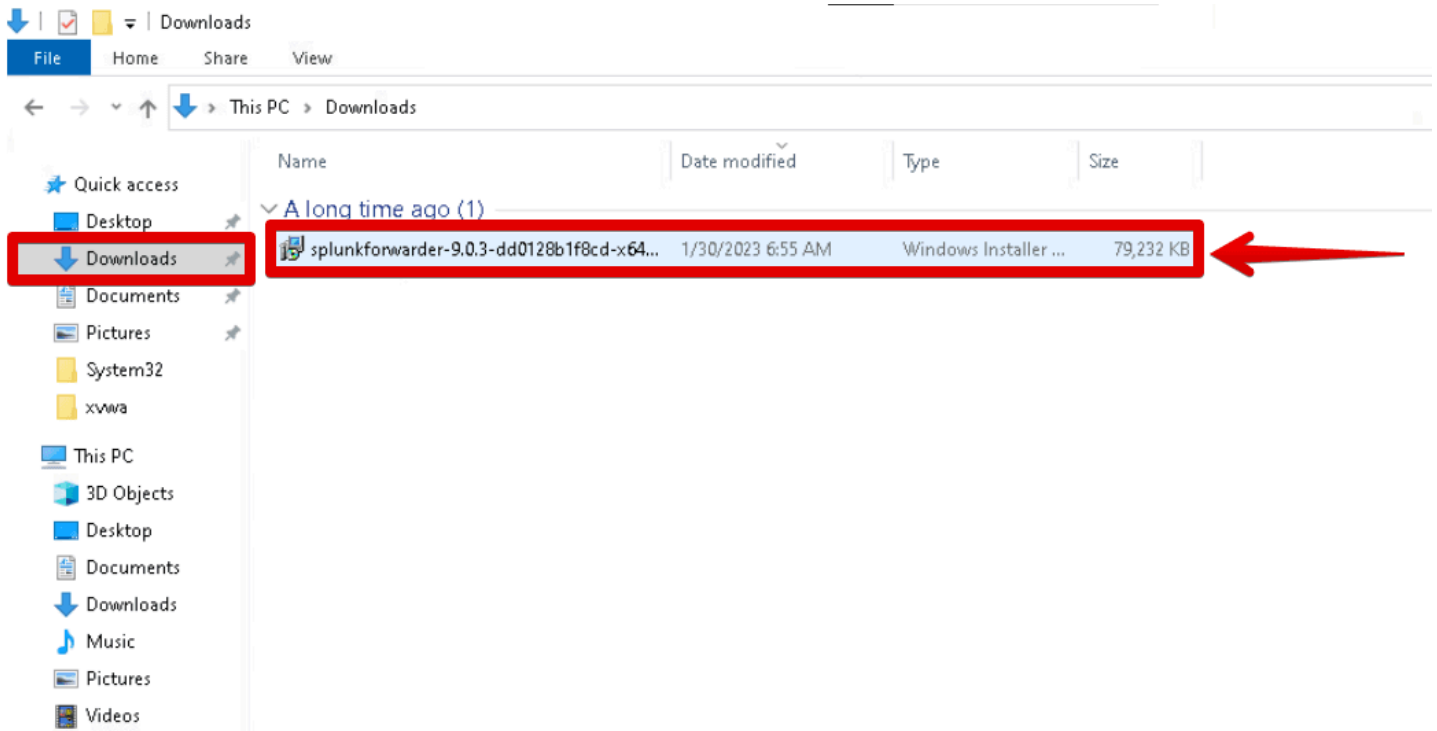
1

Connect to ACIDM01 and from the Taskbar and **open** File Explorer.



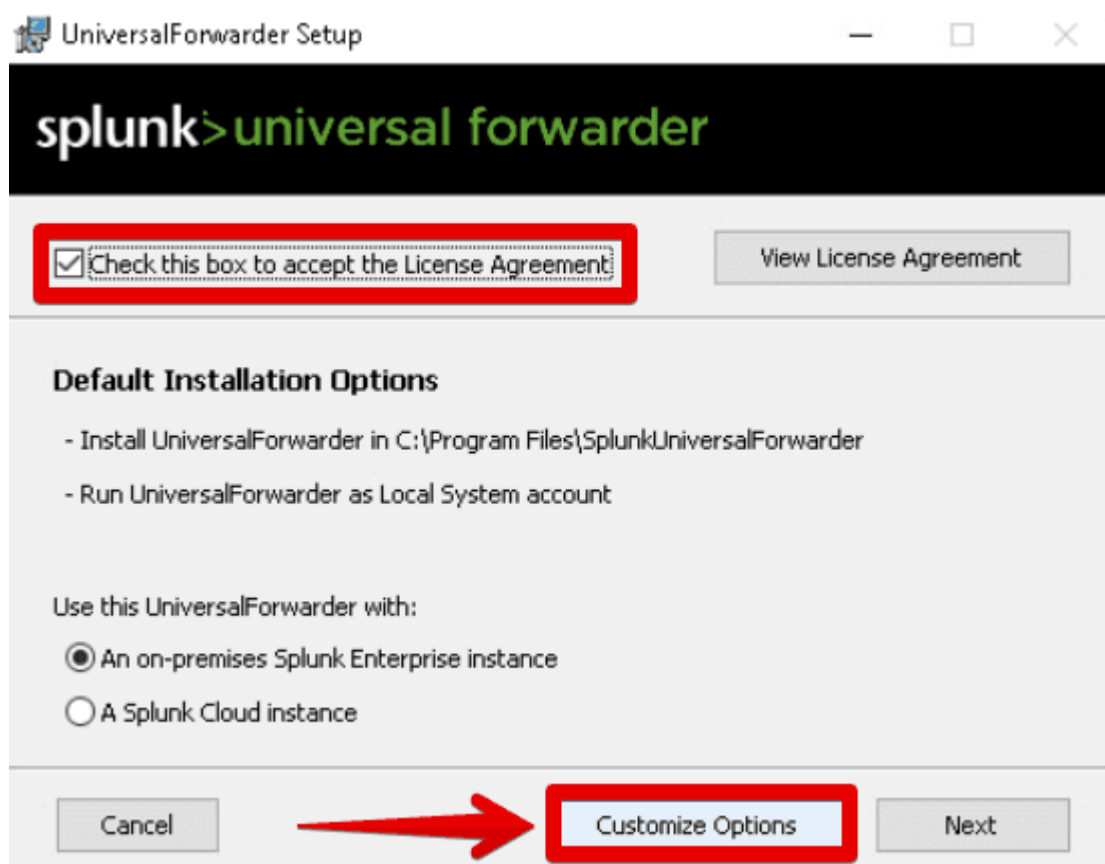
2

In **File Explorer**, **select** the **Downloads** folder and **double-click** the **splunk-forwarder-9.0.3** installation file.



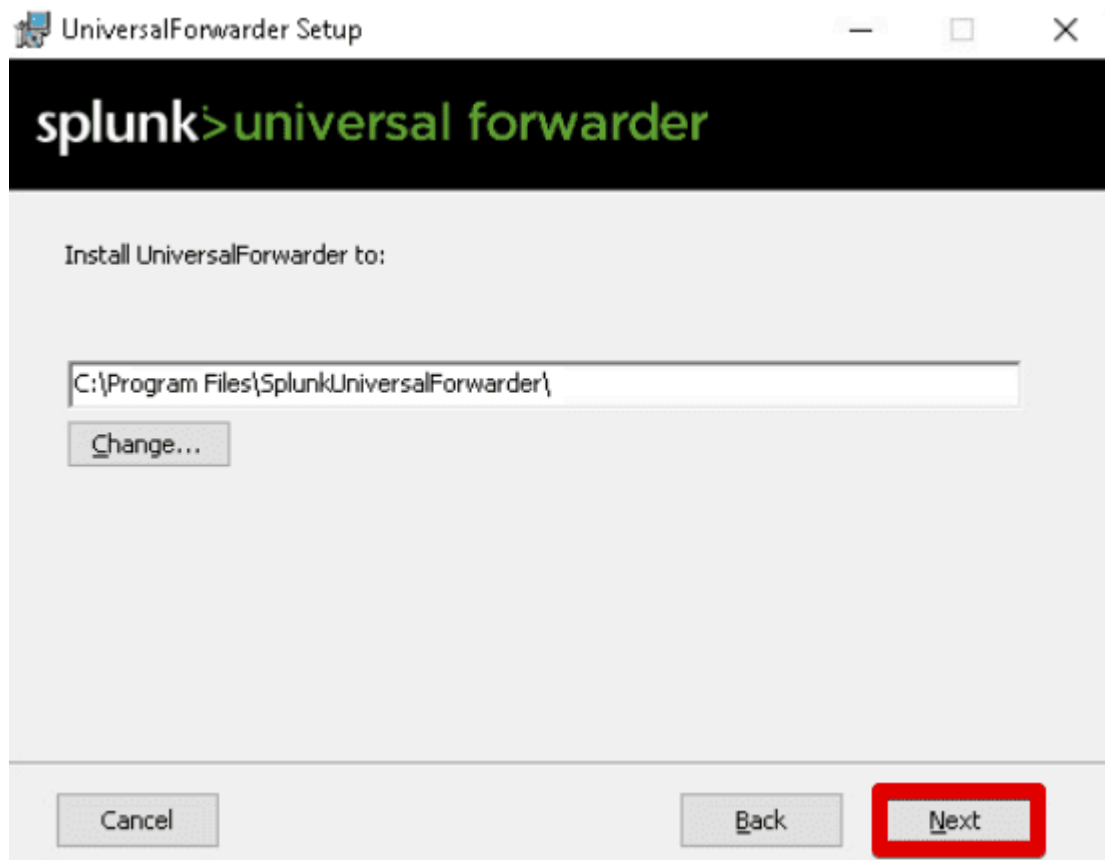
3

Tick the **Check this box to accept the License Agreement** tick box and **click** **Customize Options**.



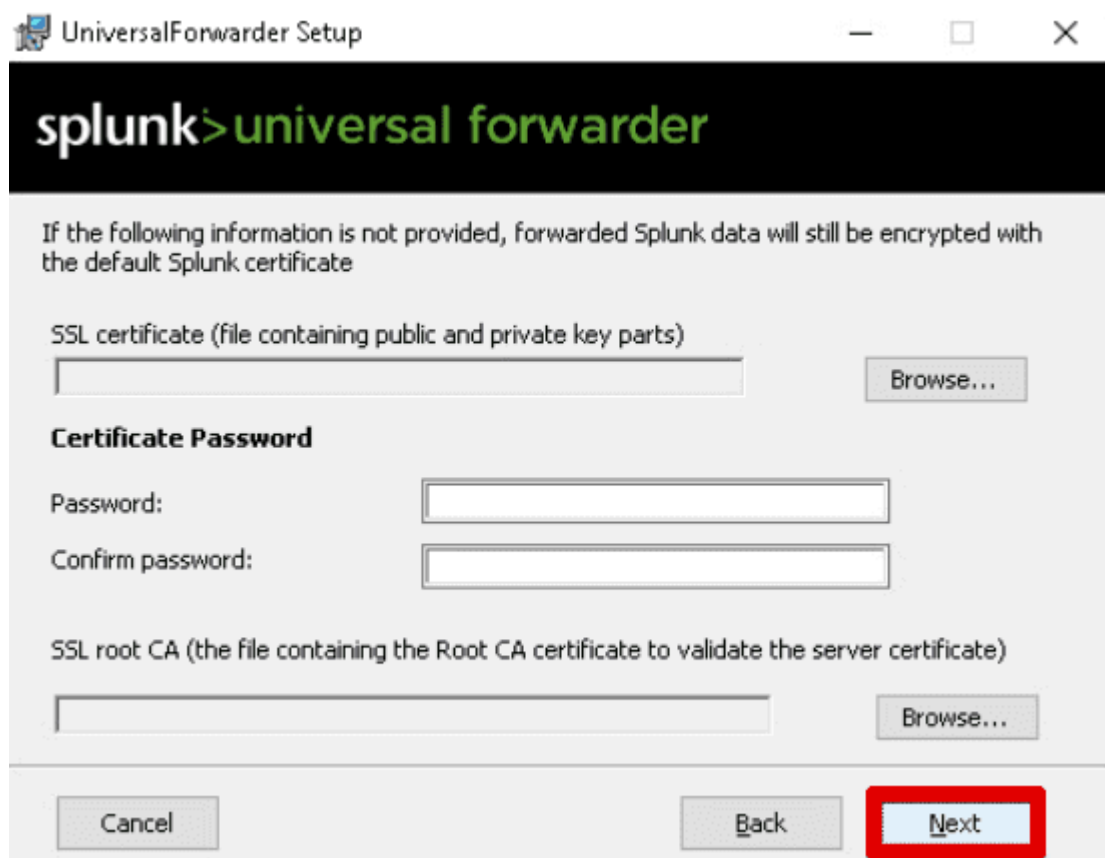
4

Click **Next** on the UniversalForwarder Setup window.



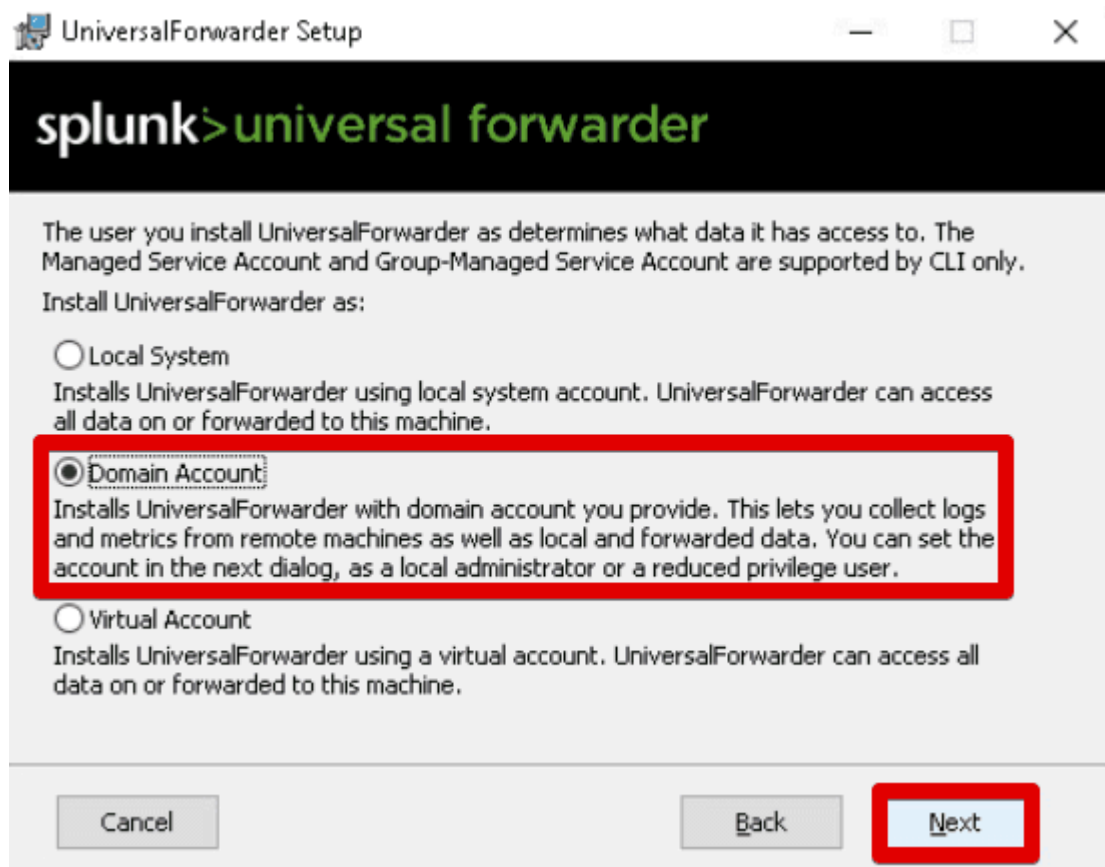
5

Click **Next** on the next installation screen.



6

Select the **Domain Account** radio button and **click** **Next**.



7

Enter the following details on the UniversalForwarder Setup screen:

Username:

Password:

Confirm password:

Click **Next**.

UniversalForwarder Setup

splunk>universal forwarder

UniversalForwarder can use Windows remote protocols like WMI to easily get data from remote machines if it has the domain rights to do so. All remote collection occurs in the user context Splunk's services are installed as, so this account must have local administrator rights as well as rights to connect and query remote machines.

Username:

Password:

Confirm password:

☒ Add user as local administrator
Leaving this box unchecked will enable reduced privilege local account. [More Info](#)

- 8 On the **UniversalForwarder Setup** screen, **tick** all the boxes in the **Windows Event Logs** pane.

Click **Next**.

UniversalForwarder Setup

splunk>universal forwarder

Windows Event Logs

- ☒ Application Logs
- ☒ Security Log
- ☒ System Log
- ☒ Forwarded Events Log
- ☒ Setup Log

Performance Monitor

- ☐ CPU Load
- ☐ Memory
- ☐ Disk Space
- ☐ Network Stats

Active Directory Monitoring
☐ Enable AD monitoring

Path to monitor

9

Untick the **Generate random password** tickbox.

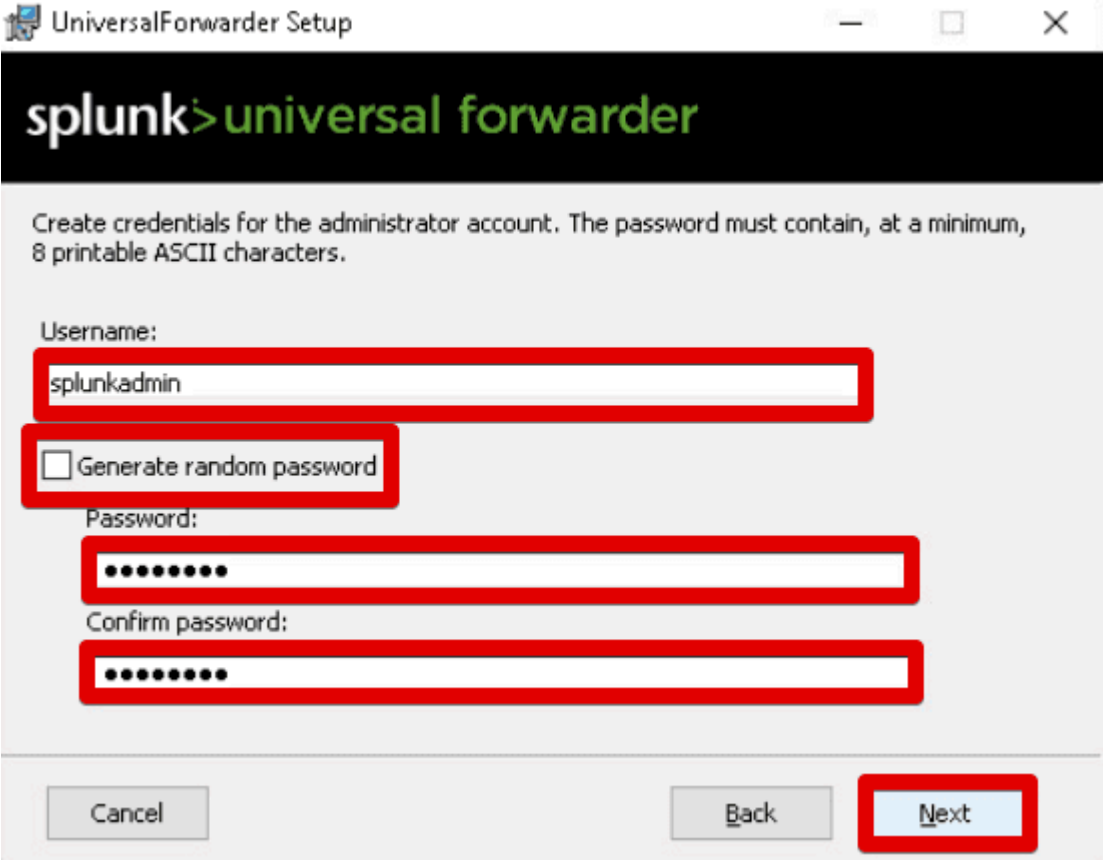
Enter the **following details** on the **UniversalForwarder Setup** screen:

Username: `splunkadmin`

Password: `Passw0rd`

Confirm password: `Passw0rd`

Click **Next**.



UniversalForwarder Setup

splunk>universal forwarder

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:
splunkadmin

☐ Generate random password

Password:
Passw0rd

Confirm password:
Passw0rd

Cancel Back Next

10

In the **Hostname or IP** field, **enter** the **following**:

`192.168.0.3`

In the **port** field, **enter** the **following**:

`8089`

Click **Next**.

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Deployment Server

Hostname or IP

192.168.0.3 : 8089

Enter the hostname or IP of your deployment server, e.g. ds.splunk.com *default is 8089*

Cancel Back Next

11

In the **Hostname or IP** field, **enter** the following:

192.168.0.3

In the **port** field, **enter** the following:

9997

Click **Next**.

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Receiving Indexer

Hostname or IP

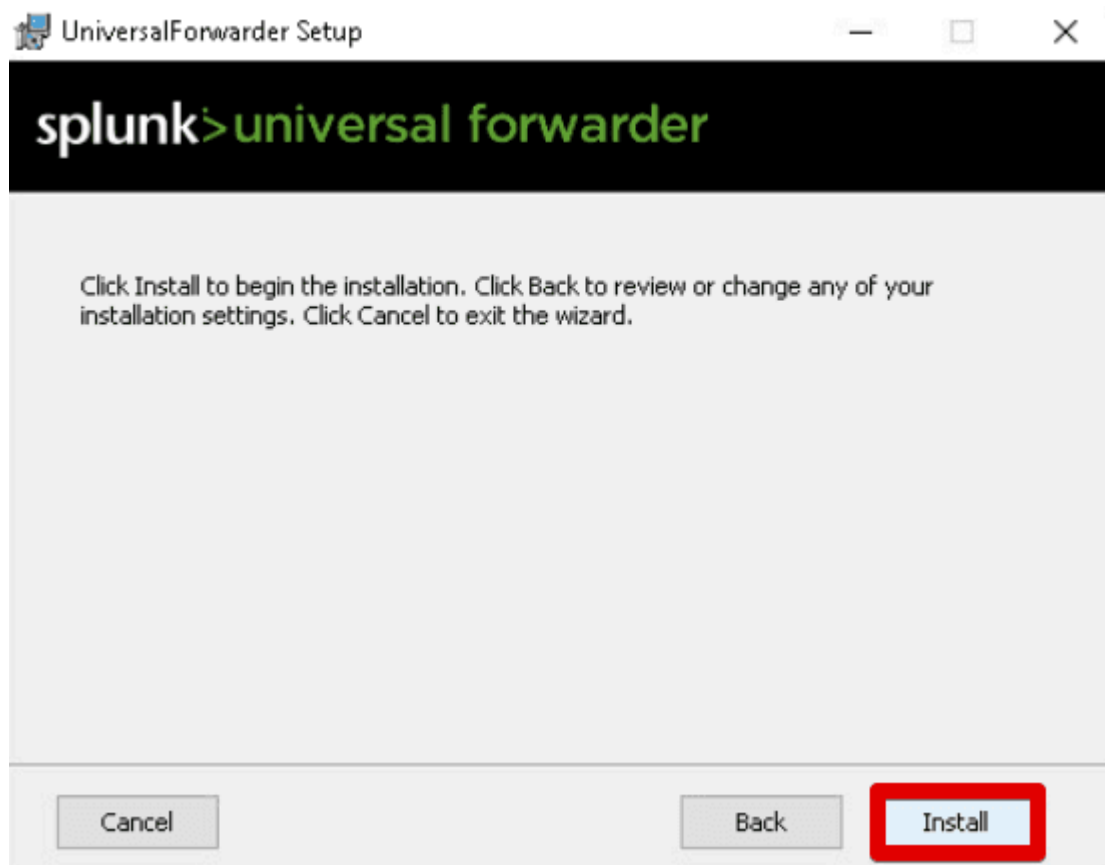
192.168.0.3 : 9997

Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com *default is 9997*

Cancel Back Next

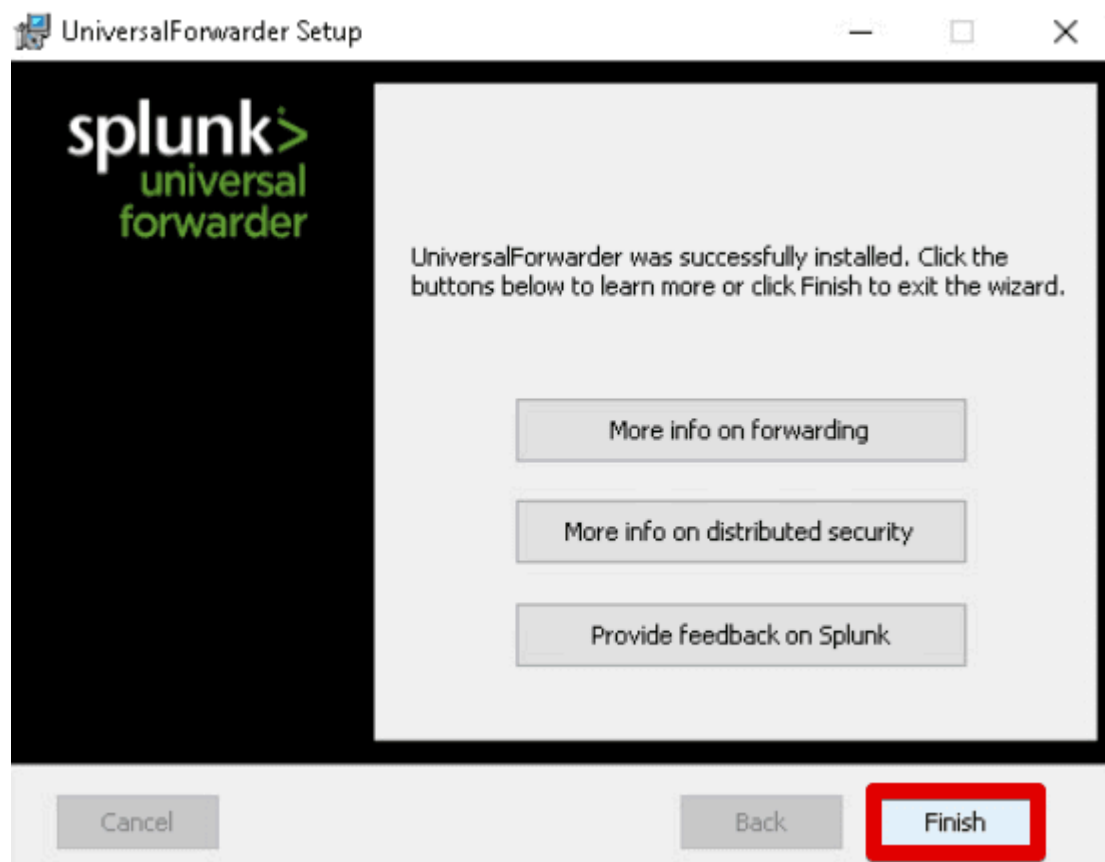
12

Click **Install** on the **UniversalForwarder Setup** window.



13

After the installation has been completed, **click Finish**.





Note: The **Splunk Universal forwarder** was configured to forward the logs of the **ACIDM01** device to the **Splunk Application**, which was configured on **ACIWIN11**. The same steps can be performed on the **ACIDC01** device to forward the logs to the **Splunk Enterprise** application.

Task 3 – Collect Logs using Splunk Enterprise

After the **Splunk Enterprise application** is installed and configured, it can be used to collect logs from the devices on the network

In this task, logs from the **ACIDM01 device** will be collected using the **Splunk Enterprise application**.

1

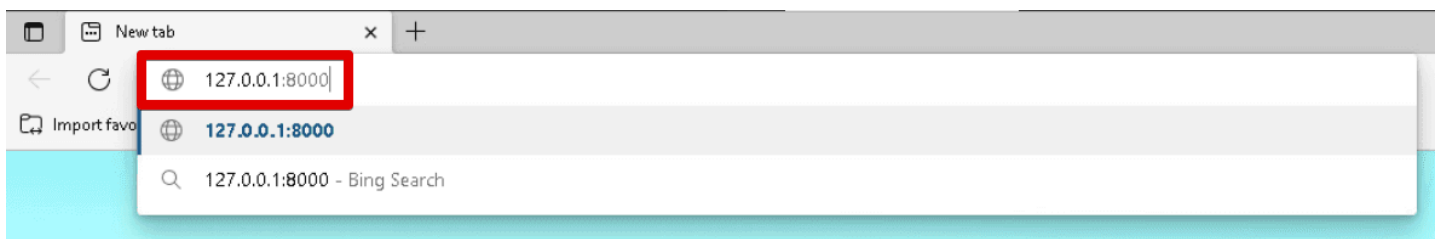
Connect to **ACIWIN11** and **open** **Microsoft Edge** from the **Taskbar**.



2

In the **Microsoft Edge browser**, **browse** to the **following URL**:

`http://127.0.0.1:8000`



3

Only if prompted, **enter** the **following credentials**:

Username: `splunkadmin`

Password: `Passw0rd`

Click **Sign In**.

4

Click Add Data in the Explore Splunk Enterprise pane.

5

If a Welcome, Administrator window pops up, **click** Skip.

6

If needed, **scroll down** on the page and **select** Forward data from a Splunk forwarder.

7

On the Add Data window, **enter** the following in the New Server Class Name field:

ACIDM01 Logs

Click WINDOWS ACIDM01 in the Available host(s) pane.

Click Next.



Note: If an error occurs, **click back** and **select Forward** again. There might be a delay before the Splunk Application detects the ACIDM01 device.

8

Click Local Event Logs on the Select Source window.

9

In **Select Event Logs** pane, **click** the following options:

Application

Security

System

Click Next.



Note: By clicking the specified **Log events**, moves it to the **Selected items** window.

10

Click **Review** on the **Input Settings** window.

11

Click **Submit** on the **Review** window.

12

Click **Start Searching** on the **Local event logs** input has been created successfully window.

13

Close the **Microsoft Edge browser** window once you have viewed the **event logs** for the **ACIDM01**.



Note: The **Event logs** for the **ACIDM01** have been successfully imported into the **Splunk Enterprise application**. Additional devices can be added to have a central repository to view all the log files in one place.

Exercise 2 – Encrypting Sensitive Data

As a **Cyber Security Specialist**, it is critical to protect sensitive data that is stored locally or in the cloud. Different types of sensitive data can be stored on the server, for example, **Personally identifiable information (PII)** or **Cardholder Data (CHD)**.

To protect locally stored data, the server's hard drives can be encrypted to protect the data on the drive.

In this exercise, the **Bitlocker Drive encryption** feature will be installed, and a local drive will be encrypted.

Learning Outcomes

After completing this exercise, you should be able to:

- Install the Bitlocker Drive Encryption Feature
- Encrypt a Local Drive on a Server

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIDM01** – Windows Server 2022 – Domain Member Server

Task 1 – Install the Bitlocker Drive Encryption Feature

To be able to encrypt a local disk on a **Windows Server**, the **Bitlocker** feature needs to be installed.

In this task, the **Bitlocker Drive Encryption** feature will be installed.

Connect to **ACIDM01**.

Click **Send Ctrl+Alt+Delete button** on the top of the screen.

Username: **ACIPLAB\Administrator**

Password: **Passw0rd**

1

In **Server Manager** and **click** **Add roles and features**.

2

In the **Add Roles and Features Wizard** window, **click** **Next**.

3

On the **Select installation type pane**, **click** **Next** until reaching the **Select features pane**.

4

On the **Select features pane**, **enable** the **Bitlocker Drive Encryption** **checkbox** and **click** **Add Features** in the pop-up window.



Note: When installing a specific feature on **Windows Server**, additional components are needed for the feature's functionality. **Windows Server** adds the additional needed components automatically.

5

Click **Next** on the **Select features pane**.

6

On the **Confirm installation selections pane**, **click** **Install**.



Note: The installation of the **Bitlocker Drive Encryption** feature will take a couple of minutes to complete.

7

Click **Close** after the installation has been completed.



Note: After the feature has been installed, the **ACIDM01 device** must be restarted to complete the installation.

8

Right-click **Start**, **select** **Shut down or sign out** and **click** **Restart**.

9

On the **Choose a reason that best describes why you want to shut down this computer** pop-up window in the drop-down menu, **select** **Other (Planned)** and **click** **Continue**.



Note: The **ACIDM01 device** will take a couple of minutes to restart to configure the installed feature.

Task 2 – Encrypting a Local Drive on a Server

After installing the **Bitlocker Drive Encryption** feature, it can encrypt and secure a local drive.

In this task, a local drive will be encrypted.

1

Reconnect to **ACIDM01** and **open** **File Explorer** from the **Taskbar**.

2

In **File Explorer**, **expand** **This PC**.

3 In the **This PC** pane, **right-click** **New Volume (D:)** and **select** **Turn on BitLocker**.

4 On the **BitLocker Drive Encryption** window, **tick** the **Use a password to unlock this drive** **checkbox** and **enter** the following:

Enter your password:

Re-enter your password:

Click **Next**.

5 On the **How do you want to back up your recovery key?** window, **select** **Save to a file**.

In the **Save BitLocker recovery key as** pop-up window, **select** **Documents** and **click** **Save**.

6 **Click** **Next** on the **How do you want to back up your recovery key?** pane.



Note: In a real-world scenario, the recovery key will not be stored on the local machine but would rather be saved to a secure network location or to a USB flash drive. This will ensure that the recovery key will be secure and accessible.

7 **Click** **Next** on the **Choose how much of your drive to encrypt** pane.



Note: As this is an empty drive, the default selection is used. In a real-world scenario, if the drive already had data on it that needs to be encrypted, the second option needs to be selected.

8

Click **Next** on the **Choose which encryption mode to use** pane.



Note: Depending on the type of drive that will be encrypted, a different option can be selected. In this case, a fixed drive will be encrypted.

9

On the **Are you ready to encrypt this drive?** pane, **click** **Start encrypting**.

10

Close **File Explorer**.



Note: In **File Explorer**, the **New Volume (D:)** icon has changed to a lock, indicating that the drive has successfully been encrypted.

Exercise 3 – Enable Multifactor Authentication

Multifactor Authentication can be enabled to add an additional layer of security to protect user accounts. Several different methods can be used to implement multifactor authentication. These include smart cards, using an authentication application or a secure USB key.

Multifactor Authentication is used in conjunction with the user's password to ensure a secure login.

In this exercise, a user will be created, and multifactor authentication will be enabled.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a User in Active Directory
- Enable Multifactor Authentication for an Active Directory User

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIWIN11** – Windows 11 – Domain Member Workstation

Task 1 – Enable Multifactor Authentication

User accounts in **Active Directory** can be secured by enabling multifactor authentication for the accounts.

In this task, a user will be created, and **Multifactor Authentication** will be enabled.

1 **Connect** to **ACIDC01** in **Server Manager**, **click** **Tools** and **select** **Active Directory Users and Computers**.

2 In the **Active Directory Users and Computers** pane on the left **expand** **ACIPLAB.COM** if it's not expanded. **Right-click** **Users**, **select** **New** and **click** **User**.

3 On the **New Object – User**, **complete** the **following**:

First name:

User logon name:

Click **Next**.

4

On the next screen, **complete** the following fields:

Password:

Confirm password:

Untick User must change password at next logon checkbox.

Click Next.

5

Click Finish on the New Object – User window.

6

Double-click the Users folder on the left pane. **Right-click** Louisplab and **select** Properties in the Active Directory Users and Computers window.

7

On the Louisplab Properties window, **select** the Account tab.

8

On the Louisplab Properties window in the Account options pane, **scroll down** and **tick** the Smart card is required for interactive logon tickbox.

Click OK.

Task 2 – Verifying Multifactor Authentication

In this task, **Multifactor Authentication** for the newly created user will be verified.

1

Connect to **ACIWIN11**, and **click** **Other user** on the **login screen**.

2

Enter the **following** on the **login screen**:

Username: **louisplab**

Password: **Passw0rd**

Press **Enter**.

3

Notice the **error message** when trying to log in with the user account.



Note: Due to the lab environment's physical restriction, multifactor authentication can not be further applied.



Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **System & Network Security Implementation Concepts** practice lab.

Summary

You completed the following exercises:

- Exercise 1 – Log Collection with Splunk
- Exercise 2 – Encrypting Sensitive Data
- Exercise 3 – Enable Multifactor Authentication

You should now be able to:

- Collect Logs from Devices on the Network using the Splunk Enterprise Application
- Encrypt a Local Drive using BitLocker Drive Encryption
- Enable Multifactor Authentication

Feedback



Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.